

RDF: N-ary Relations

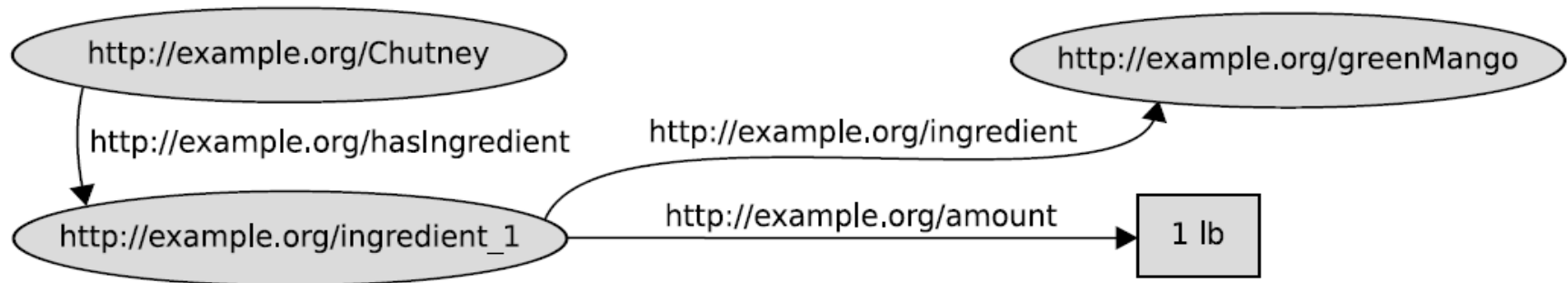
Example:

```
@prefix ex: <http://example.org/> .  
ex:Chutney    ex:ingredient    ex:greenMango;      ex:amount    "1lb" ;  
              ex:ingredient    ex:CayennePepper;  ex:amount    "1tsp." .
```



RDF: N-ary Relations

Example:



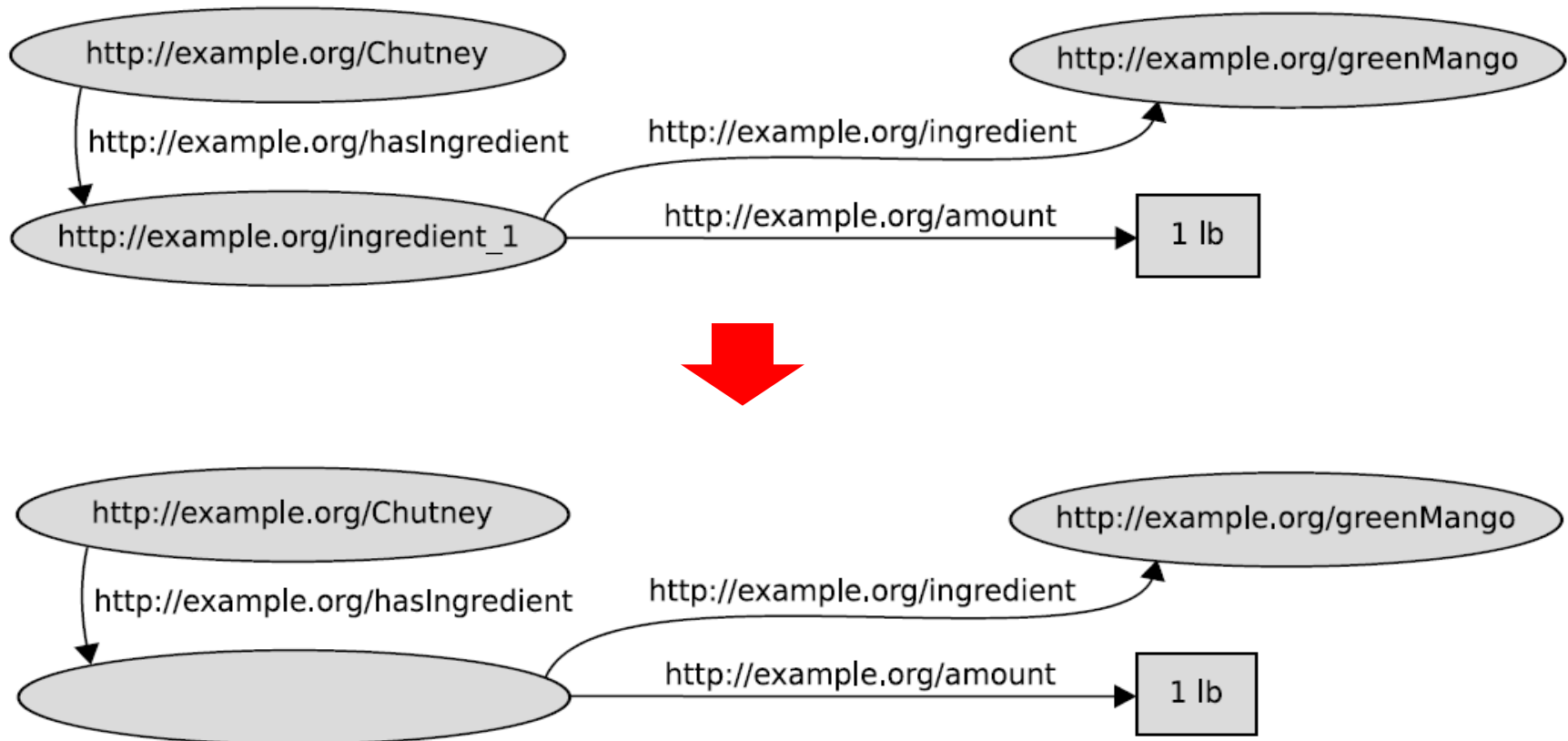
```
@prefix ex: <http://example.org/> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
ex:Chutney      ex:hasIngredient  ex:ingredient1 .
ex:ingredient1  rdf:value        ex:greenMango;
                ex:amount        "1lb" .
```

RDF: N-ary Relations

- Not all relations are binary.
- All N-ary relations can be “encoded” as a set of binary relations using auxiliary nodes, and this process is called reification.

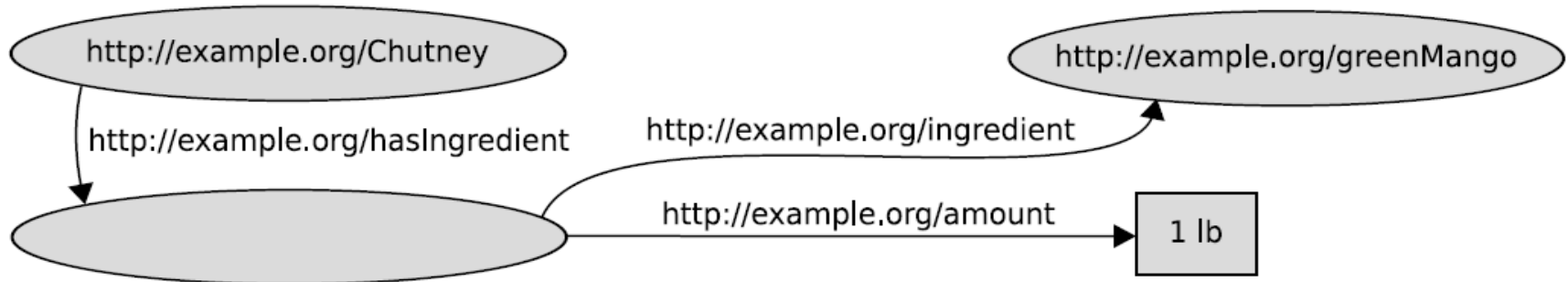
RDF: N-ary Relations and Blank Node

Example:



RDF: N-ary Relations and Blank Node

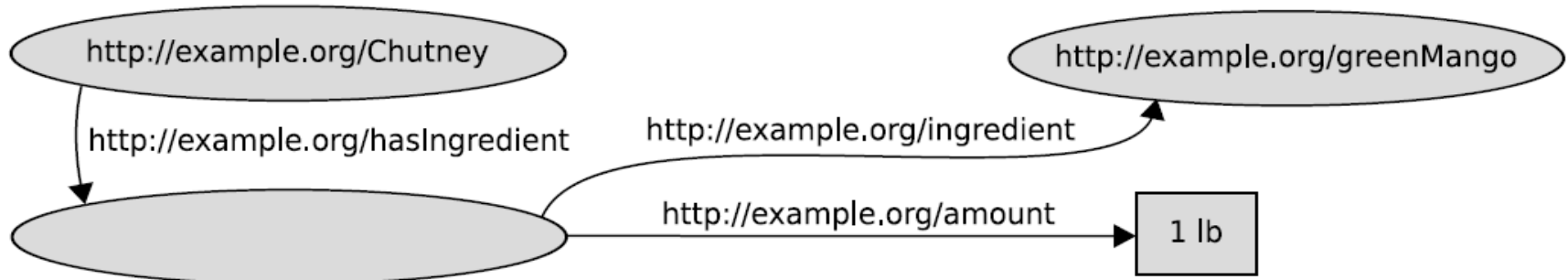
Example:



```
<rdf:Description rdf:about="http://example.org/Chutney">
  <ex:hasIngredient rdf:nodeID="id1" />
</rdf:Description>
<rdf:Description rdf:nodeID="id1">
  <ex:ingredient rdf:resource="http://example.org/greenMango" />
  <ex:amount>1lb</ex:amount>
</rdf:Description>
```

RDF: N-ary Relations and Blank Node

Example:



Shortcut:

```
<rdf:Description rdf:about="http://example.org/Chutney">
  <ex:hasIngredient rdf:parseType="Resource">
    <ex:ingredient rdf:resource="http://example.org/greenMango" />
    <ex:amount>1lb</ex:amount>
  </ex:hasIngredient>
</rdf:Description>
```

RDF VS. XML

- IRIs solve the problem of term meaning.
- Triple-based data model describe relations or properties among terms.